

Metamorphic Active Perceptive (MAP) Hand

驱感一体化刚柔耦合变胞夹爪

Key words: In-hand manipulation, adaptive grasping, ...

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Outline V1

A Synergistic Robotic Hand with a Perceptive Morphing Palm for Robust Adaptive Grasping - 一种用于鲁棒自适应抓握的具有感知变形手掌的协同机器人手

Challenges

- Passive adaption in grasping, relying on soft materials or underactuated linkages
- 3D morphological perception for dexterous manipulation, large-area sensing and 3D sensing, complex wiring for electric signals
- Pre-defined motion trajectories resulting in confined workspace and grasping envelope for existing actuated finger, requiring integrating 3D sensing and actuation and In-hand active exploration

Main contributions

- Hardware Layer: The Active-Perceptive Morphing Palm
- System Layer: A Novel Finger-Palm Synergistic Architecture
- Theory Layer: A New Active-Modulation Grasp Analysis Paradigm
- Application Layer: Physics-Informed Learning & Advanced Manipulation

Innovations

- 3D perception: 3 bending sensors for finger proprioception + 3D geometrical reconstruction for palm exteroception
- Integrated actuation: hydraulically actuated, active morphing palm (active DoF) with cable-driven flexible fingers for adaptive enveloping grasping and augmenting the stability and range of object geometries
- Synergistic operation: a new paradigm of synergistic adaptive grasping based on perception and actuation for object recognition and operation

Advantages

- Palm: morphing + 3D perception
- Finger: adaptive grasping + embedded curvature sensing
- Robotic hand: integrated sensing and actuation

Design

- Palm: diameter 80 - 100 mm, thickness 120 mm $\rightarrow$ 3 modules with diameter $\sim$ 50mm, thickness $\sim$ 40mm
- Structure: liquid tank sealed with a stretchable membrane
Morphing structure: hydraulic actuation of the membrane for shape morphing by a peristaltic pump
- Sensing principle: continuous spectral filtering with a miniature camera
- Fingers:
Bending sensor: frustrated total internal reflection
- Integration:
Fabrication and assembly: palm, finger with sensor, robotic hand

Characterization and analysis

- Palm actuation:
Pressure vs geometry, pressure vs block force, layer
- Palm 3D sensing:
- Finger kinematics analysis:
- Bending sensor performance and finger state estimation:
- Robotic hand workspace:
- Contact model:
(1) Moiton analysis (static)
Phase I: Caging (coarse positioning): three-point contact (form closure) of three fingers
Phase II: Active envelopment: introducing surface contact of active morphing palm (force closure)
(2) Grasp analysis (kinetics)
Active modulation of grasp wrench space: Grasping ability is determined by grasp wrench space, which can be tuned by the palm contacting
Active stiffness and damping tuning: Impedance control

Future work

- Learn contact characteristics
- Learn the optimal palm pressure required to grasp a specific object
- Grasping experiments

References

Pozzi, Maria, et al. "Actuated palms for soft robotic hands: review and perspectives." IEEE/ASME Transactions on Mechatronics 29.2 (2023): 902-912.

Lee, Jeongwon, et al. "Soft robotic palm with tunable stiffness using dual-layered particle jamming mechanism." *IEEE/ASME Transactions on Mechatronics* 26.4 (2021): 1820-1827.

Zhang, Ningbin, et al. "Soft robotic hand with tactile palm-finger coordination." *Nature Communications* 16.1 (2025): 2395.

Wang, Haihang, et al. "A novel soft robotic hand design with human-inspired soft palm: Achieving a great diversity of grasps." *IEEE Robotics & Automation Magazine* 28.2 (2021): 37-49.

Lei, Ziwei, et al. "A biomimetic tactile palm for robotic object manipulation." *IEEE Robotics and Automation Letters* 7.4 (2022): 11500-11507.

Tan, Yu Heng, Wenjie Li, and Huichan Zhao. "A center-less quadrotor design with a soft enveloping grasper for aerial grasping and delivery tasks." *The International Journal of Robotics Research* (2025): 02783649251339772.

Zhang, Ningbin, et al. "Biomimetic rigid- soft finger design for highly dexterous and adaptive robotic hands." *Science Advances* 11.17 (2025): eadu2018.

Outline V2

Challenges

- Functional separation of sensing and actuation, lacking active zooming capabilities
- Passive support of the central palm
- Palm-finger coordination for precision recognition and manipulation

A core challenge in robotic manipulation is the functional passivity of the central palm, which remains a static support rather than an active, perception-integrated system.

Our work facilitates a paradigm shift from passive structural adaptation to active closed-loop manipulation within a hybrid perception-integrated palm-finger structure.

Main contributions

- *Perception-actuation integration*: Integration and decoupling of high-resolution topographic sensing and hydraulic actuation in a fluidic interface (Active zooming tactile sensing; decoupling).
- *Embodied morphological reconfiguration*: An adaptive palm with integrated perception and actuation via centripetal-driven morphological reconfiguration (Centripetal mechanism and three active palm interfaces).
- *Hierarchical palm-finger coordination*: Hierarchical palm-finger coordination paradigm for robust grasping, multi-scale recognition, and precise in-hand manipulation.

Structural design

1. Perceptive morphing palm
 - Perception-actuation integrated module
 - Centripetal mechanism with a flexible fringe
2. Flexible finger
 - Tendon-driven finger
 - Optical waveguide for bending angle and contact force analysis

Characterization and analysis

1. Perception-actuation integrated module
 - Static state: Spatial accuracy and resolution
 - Dynamic state: Volume vs deflection, active zooming tactile resolution, sensitivity and measurement range
 - 3D reconstruction of various objects
2. Centripetal mechanism a flexible fringe
 - Kinematic analysis (motion coupling)
 - Operational workspace analysis
3. Tendon-driven fingers
 - Modeling
 - Kinematic Workspace
 - Output force

4. Optical waveguides
 - Bending angle analysis
 - Contact force analysis

Demonstrations

1. Cross-scale high-fidelity recognition from geometry to texture (finger for fixing and palm for active perception)

Spherical geometry: Glass ball, golf ball, tennis ball, balloon, water ball, mini basketball, lemon, grape (Stiffness, weight, geometry, texture).
2. Grasping and recognizing diverse objects with unstructured geometries

Screwdriver, wrench, bottled water, potato chips, tissue, fabric, strawberry, cards, electric drill, collaborative grasping and multi-object grasping
3. Perception-guided precision in-hand manipulation and reconstruction (closed-loop control)

In-hand rotation of a dice; Adjusting liquid level in a graduated flask/a metal ball in an enclosed platform

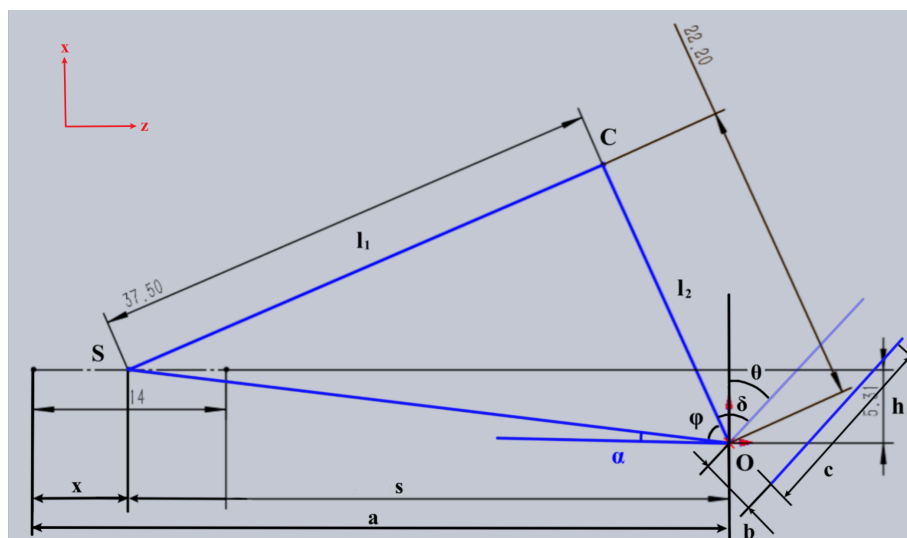
Applications

The "*sense-grasp-inspection-assemble*" pipeline for industrial scenarios

- Stage 1: Scanning for reconstruction
- Stage 2: Decision-making for grasping strategy and grasping
- Stage 3: Inspection of micro-structures
- Stage 4: Assemble

1 Kinematic Modeling of the Morphing Palm

由于手掌的三个模块呈圆周对称（ 120° 均匀分布），要计算传感平面的运动空间，可以先针对其中一个模块进行建模，不难发现其本质为一以滑块为主动件的**摇杆滑块机构**。先做出如图定义：



- 原点 O ：定轴转动点（图中红色坐标系原点）
- 滑块铰链点 S ：滑块与连杆连接点
- 摇杆铰链点 C ：摇杆与连杆连接点
- 水平轴（三维模型中的 z 轴）：向左为负，向右为正
- 竖直轴（不妨假设为三维模型中的 x 轴）：向上为正，向下为负
- 水平偏移 x ：直线电缸的位移，与滑块铰链点与原点的最大水平距离 a ，以及滑块铰链点与原点 O 在水平方向的距离 s 的关系为 $x + s = a$ ；为方便起见，暂时将 s 视为输入量
- 输出量 θ ：传感面与图中竖直面的夹角
- 常量定义：
 - 连杆长度 $l_1 = 37.50 \text{ mm}$
 - 摇杆长度 $l_2 = 22.20 \text{ mm}$
 - 偏置高度 $h = 5.31 \text{ mm}$
 - 滑块铰链点与原点的最大水平距离 $a = 50.5 \text{ mm}$
 - 传感面与摇杆平面夹角 $\delta = 63.23^\circ$
 - 传感面距离原点的距离 $b = 24.18 \text{ mm}$
 - 传感面最远端距离原点沿该平面的距离 $c = 31 \text{ mm}$

1.1 建立单模块的运动学映射 $z \rightarrow \theta$

1. 首先，计算OS的长度以及辅助角 α ：

$$\overline{OS} = \sqrt{s^2 + h^2} \quad (1)$$

$$\alpha = \arctan\left(\frac{h}{s}\right) \quad (2)$$

2. 利用余弦定理求内角 φ ：

在 $\triangle OSC$ 中，有：

$$l_1^2 = l_2^2 + \overline{OS}^2 - 2l_2 \cdot \overline{OS} \cdot \cos\varphi$$

于是：

$$\varphi = \arccos\left(\frac{l_2^2 + \overline{OS}^2 - l_1^2}{2l_2 \cdot \overline{OS}}\right) \quad (3)$$

3. 得出最终角度 θ ：

由图中几何关系可知： $\theta = \alpha + \varphi + \delta - 90^\circ$ ，代入(1)(2)(3)有：

$$\theta(s) = \arctan\left(\frac{h}{s}\right) + \arccos\left(\frac{l_2^2 + s^2 + h^2 - l_1^2}{2l_2\sqrt{s^2 + h^2}}\right) + \delta - 90^\circ \quad (4)$$

1.2 确定手掌传感面末端的局部坐标（二维）

针对每个传感模块，可以通过五条直线围成的区域范围在 $O_i - xy$ 坐标系下描述传感平面的形状：

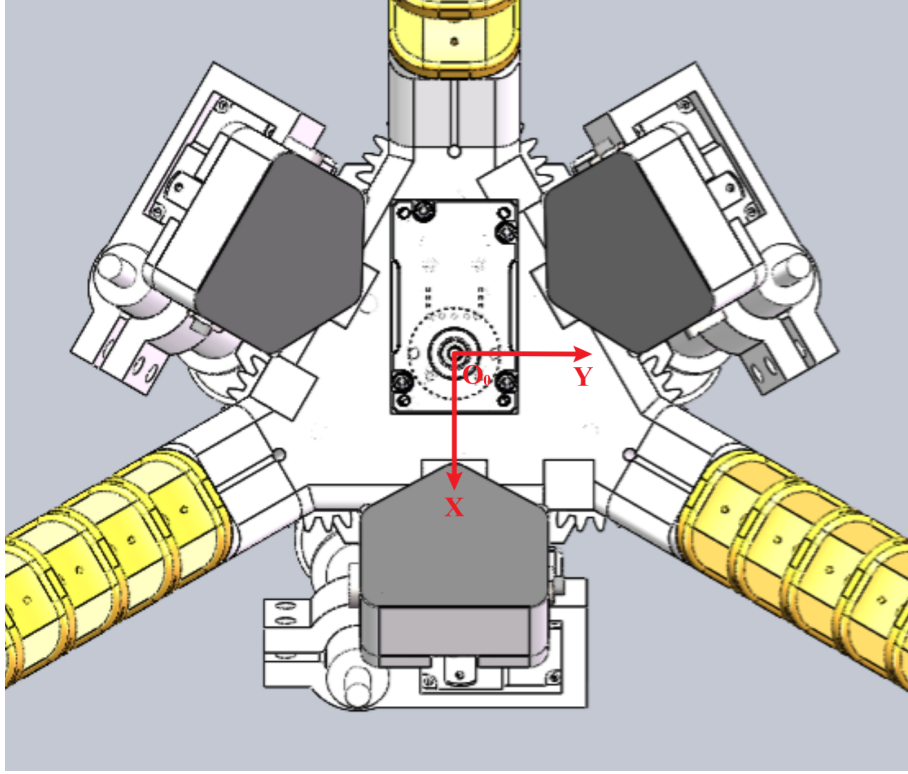
$$\begin{cases} y = 15 \\ y = -15 \\ x = 6 + \frac{\sqrt{3}}{3}y \\ x = 6 - \frac{\sqrt{3}}{3}y \\ x = 31 \end{cases} \quad (5)$$

该平面在 $O_i - xyz$ 坐标系中的真实位置为(5)中描述的平面先向 z 轴正方向移动 b ，再绕 y 轴顺时针旋转 θ° 。该变换 T 可表示为：

$$T = \begin{bmatrix} \cos\theta & 0 & -\sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & b \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta & -b \cdot \sin\theta \\ 0 & 1 & 0 & 0 \\ \sin\theta & 0 & \cos\theta & b \cdot \cos\theta \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

1.3 确定手掌传感面的工作空间（三维）

手掌三个传感模块呈 120° 分布，俯视视角如下图：



图中全局坐标系记为 $O_0 - XYZ$ ，每个传感模块的坐标系原点改记为 $O_i (i = 1, 2, 3)$ 。

为了将三个模块统一到全局坐标系，我们需要两个变换步骤：

1. 局部平移变换 T_{base}

将连杆坐标沿 X 轴正向平移基座半径 $R_{\text{base}} = 20.5\text{mm}$ ；

2. 全局坐标变换 $R_z(\gamma_i)$

将第 i 个模块绕全局 Z 轴旋转 γ_i ，其中 $\gamma_i \in \{0^\circ, 120^\circ, 240^\circ\}$ 。

综合变换矩阵 T_i 为：

$$T_i = R_z(\gamma_i) \cdot T_{\text{base}} = \begin{bmatrix} \cos \gamma_i & -\sin \gamma_i & 0 & 0 \\ \sin \gamma_i & \cos \gamma_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & R_{\text{base}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \gamma_i & -\sin \gamma_i & 0 & R_{\text{base}} \cos \gamma_i \\ \sin \gamma_i & \cos \gamma_i & 0 & R_{\text{base}} \sin \gamma_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

因此，联立(6)(7)可得，(5)中由点集 $\{(x, y)\}$ 描述的平面到全局坐标系的最终映射公式为：

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = T \cdot T_i \cdot \begin{bmatrix} x \\ y \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & -\sin \theta & -b \cdot \sin \theta \\ 0 & 1 & 0 & 0 \\ \sin \theta & 0 & \cos \theta & b \cdot \cos \theta \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \gamma_i & -\sin \gamma_i & 0 & R_{\text{base}} \cos \gamma_i \\ \sin \gamma_i & \cos \gamma_i & 0 & R_{\text{base}} \sin \gamma_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 0 \\ 1 \end{bmatrix}$$

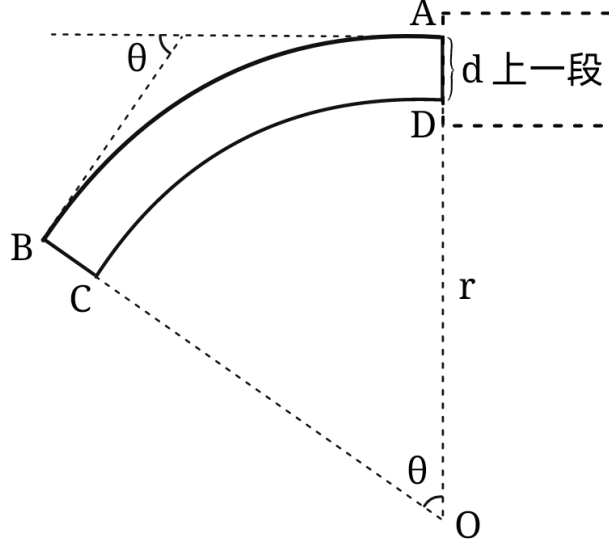
其中 θ 由(4)定义， $\gamma_i \in \{0^\circ, 120^\circ, 240^\circ\}$ 。

2 Kinematic Modeling of the Flexible finger

【弃用，存在问题】

2.1 建立腱绳长度与手指末端弯曲角度的运动学映射 $l \rightarrow \theta$

分段等曲率假设下，考虑某段梁 i 末端平面相比初始状态的弯曲角度 θ_i ：



- 原点 O_i ：该段梁弯曲的曲率中心
- $\widehat{A_i B_i}$ ：梁中性面在视图平面的投影，确定后该值不可变
- $\widehat{C_i D_i}$ ：腱绳长度，可随绕线而缩短（初始为与 $\widehat{A_i B_i}$ 等长）；当曲率较小时，可近似为 $\widehat{C_i D_i}$
- d ：梁中性面与腱绳之间的距离（恒定）
- r_i ：该段梁的曲率半径，随腱绳的长度变化而变化

梁 i 的腱绳从初始位置缩短 l_i ，有：

$$\begin{aligned} l_i &= \widehat{A_i B_i} - \widehat{C_i D_i} \\ &= \theta_i \cdot (r_i + d) - \theta_i \cdot r_i \\ &= \theta_i \cdot d \end{aligned} \quad (8)$$

设总共有 n 段这样的梁串联，则腱绳总收缩长度 l 与串联后的梁总弯曲角度 θ 满足：

$$l = \sum_{i=1}^n l_i = \sum_{i=1}^n \theta_i \cdot d = d \cdot \sum_{i=1}^n \theta_i = \theta \cdot d$$

对时间求导，有：

$$v = \omega \cdot d \quad (9)$$

其中 v 为舵机绕线的线速度， ω 为手指弯曲角度变化的速度。因此若手指建模采用分段等曲率假设，在一定的近似下，手指的弯曲角度与腱绳收缩长度之间存在固定的关系，与手指长度无关。

但是仅依靠此关系无法得知手指的具体形状，因为每一段弯曲的具体角度 θ_i 未知，需要进一步进行动力学分析。

3 Lateral Stiffness of Flexible Finger (Cantilever Beam)

已知悬臂梁的刚度公式如下：

$$k = \frac{F}{\delta} = \frac{3EI}{L^3}$$

其中 F 为负载（N）， δ 为挠度（mm）， E 为杨氏模量， I 为截面二次力矩， L 为力臂。

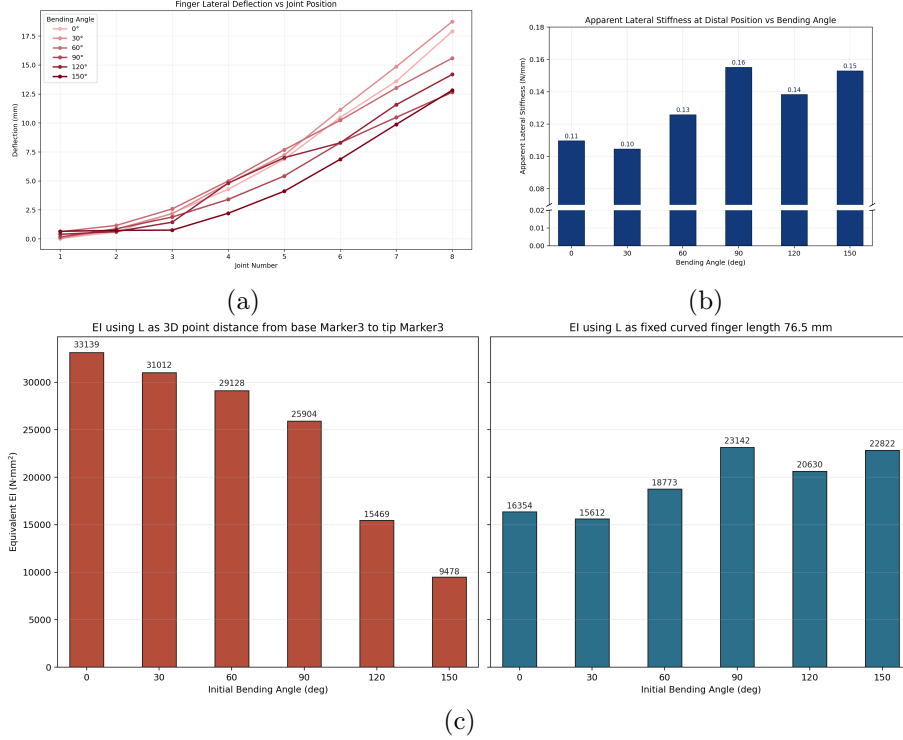


图 1. 手指侧向刚度表征

(a) 不同角度、手指不同指节挂相同重物的挠度；(b) 不同角度，手指末端挂重物的表观刚度；(c) 不同角度，手指末端挂重物的本征刚度EI，左侧使用L为实际力臂，右侧使用L为沿手指长度（常量）

结论：

根据最终图像左半部分所示结果，在以“指根 Marker3 至指尖 Marker3 的三维点距”表征等效受力臂长度的前提下，手指的等效弯曲刚度指标 EI 总体随初始弯曲角度增大而减小。具体而言，随着手指由小角度逐步过渡到大角度弯曲，指根至指尖的空间距离持续缩短，而手指本体的材料组成与结构总长度并未发生改变。这意味着，在外载恒定的条件下，系统的有效力臂减小，手指不再表现为单一均匀直梁，而更接近于多个柔性梁单元沿串联链依次参与变形的状态。对于这种串联式指节结构，可类比于弹性单元的串联效应：当相同柔性单元以更紧凑的初始构型串联工作时，整体结构对末端侧向载荷的等效抗弯能力将下降。因此，实验中观察到的 EI 随弯曲加深而降低的趋势，在力学上是可以理解且具有一致性的。

与此同时，结果也表明，当前实验并未明显呈现出“随着弯曲角度增加，因腱绳张力升高而引起整体刚度增强”的效应。换言之，在本组数据中，由几何构型变化和串联柔性累积所导致的刚度下降趋势，可能占据了主导地位，从而掩盖了腱绳预张力增加所可能带来的局部刚化贡献。因此，就现阶段结果而言，更稳妥的结论应当是：在腱绳驱动存在的实际工况下，手指侧向等效刚度主要表现出随初始弯曲程度增加而降低的总体趋势，而尚不能据此直接证明腱绳张力提升会显著增强侧向刚度。

进一步地，若希望严格识别腱绳张力对刚度的独立贡献，则需要引入对照实验，将“结构几何效应”与“驱动预张力效应”加以分离。理想方案是在尽可能保持相同初始姿态的条件下，比较有腱绳驱动与无腱绳驱动两种状态下的侧向响应差异。然而，这类实验在实施上存在明显困难：一旦去除腱绳驱动，如何稳定维持与驱动状态一致的手指弯曲姿态，本身就构成了额外的实验挑战。因此，后续若要进一步论证腱绳张力的刚化作用，仍需发展更可控的姿态保持与对照加载方法。

刚度：

如果只是为了实验比较，我觉得最合适的表征方式有两个层级：

1. 最稳妥的实验量：直接用 $k_{\text{app}} = \frac{F}{\delta}$ ，把它称为“末端表观侧向刚度”。这几乎不依赖模型假设，最安全，也最容易解释
2. 如果一定要引入“EI”：建议明确写成“基于等效悬臂假设反算得到的构型相关等效 EI”或者“equivalent EI under an Euler-Bernoulli-inspired approximation”，这样就不会把它误解成材料本征参数